

# Daniel Sánchez-García

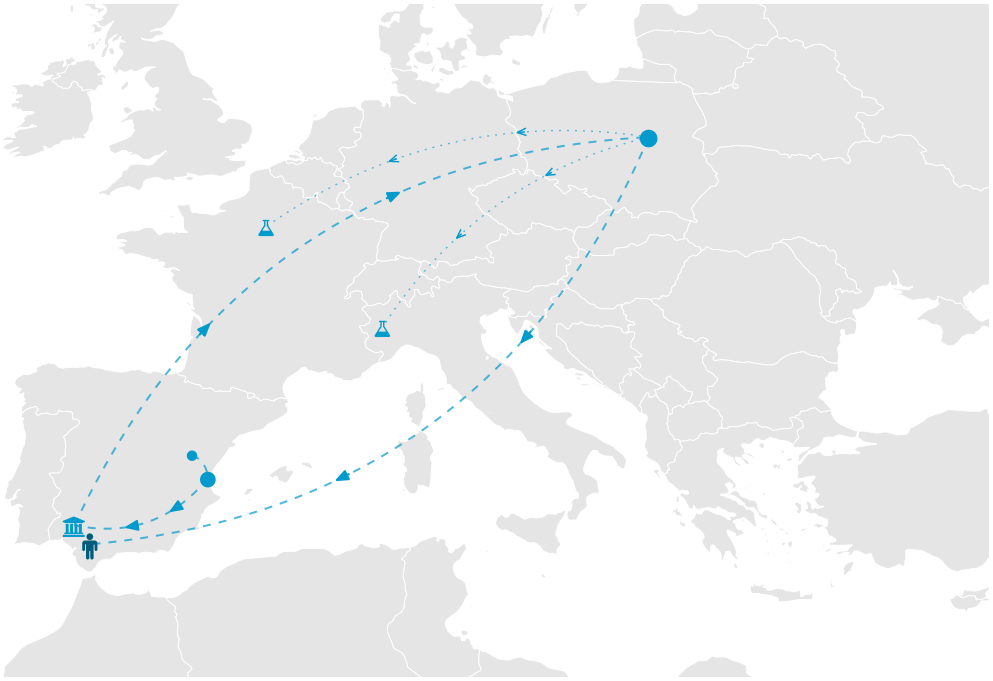
PhD in Biology

*Doñana Biological Station - EBD-CSIC - Seville (Spain)*

+34 638 546 801 | danielsangarci@gmail.com

danielsangarci.com | danielsangarci | danielsangarci | 0000-0002-0710-6292

## My journey



## Professional experience

### Postdoctoral Researcher

Doñana Biological Station (EBD-CSIC), Department of Conservation Biology and Global Change, Seville (Spain) — Jan 2025 - present

### Specialist in ant species identification

Doñana Biological Station (EBD-CSIC), Department of Conservation Biology and Global Change, Seville (Spain) — May 2024 - Jun 2024

- Identification of ant species from pitfall trap samples from Spain, France, Belgium and the Netherlands, contributing to ecological research and biodiversity assessments through taxonomic expertise.

## Education

### PhD in Biology

Museum and Institute of Zoology, Polish Academy of Sciences, Warsaw (Poland) — Oct 2019 - Sep 2024

- PhD Thesis: The project aimed to get a better understanding of how reintroductions of parasites work, we performed a multi-level comparison of source and reintroduced populations of *Phengaris teleius* by studying 1) population genetics, 2) morphology of adult butterflies and 3) behavioural, chemical and acoustical adaptations of butterfly caterpillar to the *Myrmica* host ants.

### MSc in Biodiversity and Conservation Biology

Universidad Pablo de Olavide, Seville (Spain) — Oct 2017 - Jul 2018

- MSc Thesis (Thesis grade: 9.8/10 - Honors with Distinction): The work aimed to study the effect of temperature over ant assemblages from different habitats, applying different ecological perspectives. Taxonomic and functional diversity were studied, and a new index was developed to study the direct effect of temperature over the interspecific competition.

### BSc in Biology

Universitat de València, Valencia (Spain) — Sep 2012 - Sep 2016

- BSc thesis (Thesis grade: 9.5/10): The work aimed to provide new knowledge about the ant communities of Sierra de Javalambre. The main goals were: 1) study how ant species are distributed along the altitudinal gradient, studying variables like ground temperature, nest density and stone density; 2) study the patching effect of *Juniperus sabina* over ant activity and nesting areas.

## Research stays

### Prof. Luca Pietro Casacci

University of Turin, Department of Life Sciences and Systems Biology, Turin (Italy) — Oct 2021

- I received training in recording insect vibroacoustic signals, sonogram analysis and performing behavioral assays, which enhanced my skills in experimental design and data interpretation.

### Prof. Patrizia d'Ettorre

Sorbonne Paris North University, Laboratory of Experimental and Comparative Ethology, Paris (France) — Jan 2020 - Feb 2020

- I received training in GC-MS analysis of cuticular hydrocarbons and bioinformatics workflows, which deepened my understanding of chemical communication in social insects and enhanced my analytical skills.

## Internships

### Scientific assistant

Valencian Institute of Agricultural Research (IVIA), Department of Entomology, Valencia (Spain) — Feb 2016 - May 2016

- Internship with the researcher Alejandro Tena working on a biological control applied study about the effect of the presence of the ant *Lasius grandis* on the behaviour of the parasitoid wasp *Anagyrus pseudococci* on mealybug colonies of *Planococcus citri* (pest of citrus).

### Environmental manager

Agricultural, Stockbreeding and Environmental Public Service, Government of Aragón, Teruel (Spain) — Jul 2015 - Sep 2015

- Internship with the environmental manager José Manuel González working on environmental impact reports; monitoring of fish river population dynamic for its management and conservation using electric fishing techniques; monitoring of deer population dynamic for its management and conservation in the National Hunting Reserve "Montes Universales"; management and translocation of native plant species.

## Publications

Also in *my Google Scholar profile* and available to download from *my Website*.

Ambakina, H., Gómez, K., Monzenga, J.-C., **Sánchez-García, D.**, & Dekoninck, W. (2026). Uncovering ant diversity across forest successional stages in the Yangambi Biosphere Reserve (DRC): Insights from Winkler and pitfall trap sampling. *African Invertebrates*, *67*(1), 65–91. https://doi.org/10.3897/afrinvertebr.67.180461

**Sánchez-García, D.**, Cerdá, X., & Angulo, E. (2026). Temperature and management reduce the abundance of the invasive Argentine ant (Linepithema humile). *bioRxiv*. https://doi.org/10.64898/2026.10.08.756264

**Sánchez-García, D.**, Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Ongoing coevolution between reintroduced Phengaris teleius butterflies and their Myrmica host ants. *bioRxiv*. https://doi.org/10.64898/2025.12.02.691951

**Sánchez-García, D.**, Wynhoff, I., Kajzer-Bonk, J., Sztencel-Jabłonka, A., Nowicki, P., Casacci, L. P., & Witek, M. (2024). Temporal and spatial variation of morphological traits and genetic structure in Phengaris teleius myrmecophilous butterflies following habitat and climate changes three decades after reintroduction. *Global Ecology and Conservation*, *54*, e03104. https://doi.org/10.1016/j.gecco.2024.e03104

Trigos-Peral, G., Maák, I. E., Schmid, S., Chudzik, P., Czaczkes, T. J., Witek, M., Casacci, L. P., **Sánchez-García, D.**, Lőrincz, Á., Kochanowski, M., & Heinze, J. (2024). Urban abiotic stressors drive changes in the foraging activity and colony growth of the black garden ant Lasius niger. *Science of The Total Environment*, *915*, 170157. https://doi.org/10.1016/j.scitotenv.2024.170157

**Sánchez-García, D.**, Cerdá, X., & Angulo, E. (2022). Temperature or competition: Which has more influence on Mediterranean ant communities? *PLOS ONE*, *17*(4), e0267547. https://doi.org/10.1371/journal.pone.0267547

Angulo Aguado, E., Castro Cobo, S., **Sánchez-García, D.**, Sergio, F., Reyes-López, J. L., Álvarez Blanco, P., & Cerdá Sureda, X. (2019). ¿Se podrá controlar la expansión de la invasora hormiga argentina en Doñana? In J. Junoy (Ed.), *Especies exóticas invasoras: Cátedra de parques nacionales* (pp. 249–262). Editorial Universidad de Alcalá.

García, F., Cuesta-Segura, A., Espadaler, X., García, J. C., & **Sánchez-García, D.** (2019). Lasius piliferus Seifert, 1992: descripción de la reina y actualización de su distribución ibérica (Hymenoptera: Formicidae). *Boletín de La Sociedad Entomológica Aragonesa (S.E.A.)*, *65*, 39–44.

García García, F., Espadaler Gelabert, X., Cuesta-Segura, A. D., & **Sánchez-García, D.** (2018). Primera cita ibérica para Temnothorax conatensis Galkowski & Lebas, 2016, y actualización de la distribución para Temnothorax grouvellei (Bondroit, 1918) (Hymenoptera: Formicidae). *Iberomyrmex*, *10*, 22–27.

Espadaler, X., **Sánchez-García, D.**, & García-García, F. (2017). Temnothorax ibericus Menozzi (1922), un endemismo ibérico orófilo (Hymenoptera, Formicidae). *Iberomyrmex*, *9*, 5–9.

**Sánchez-García, D.**, & Espadaler, X. (2017). Una nueva especie parásita social para la península ibérica. Bothriomyrmex communista Santschi, 1919 (Hymenoptera: Formicidae) en España. *Iberomyrmex*, *9*, 11–13.

**Sánchez-García, D.**, & Espadaler, X. (2015). Cardiocondyla obscurior Wheeler, 1929 (Hymenoptera, Formicidae) en España. *Iberomyrmex*, *7*, 7–9.

## Conference talks and posters

**Sánchez-García, D.**, Stefanini, I., Lőrincz, Á., Trigos-Peral, G., Casacci, L. P., Ślipiński, P., Maák, I. E., Malewski, T., Oszako, T., & Witek, M. (2026). Ants shape the phyllosphere microbiome and drive tree health. *XX Iberian Congress of Myrmecology, Taxomara 2026*, *Móstoles*, Spain (Poster).

**Sánchez-García, D.**, Taheri, A., Cerdá, X., & Angulo, E. (2025). ANT HUNTERS: A citizen science initiative for ant biodiversity research. *XIX Congress of the Spanish Society of Ethology and Evolutionary Ecology*, *Seville*, Spain (Poster).

**Sánchez-García, D.**, Taheri, A., Cerdá, X., & Angulo, E. (2025). ANT HUNTERS: La ciudadanía a la caza de la biodiversidad de hormigas. *XIX International Congress of Myrmecology, Taxomara 2025*, *Girona*, Spain (Poster).

**Sánchez-García, D.**, Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Mariposas mirmecófilas reescriben su lenguaje para adaptarse a nuevas hormigas huésped. *XIX International Congress of Myrmecology, Taxomara 2025*, *Girona*, Spain (Talk).

**Sánchez-García, D.**, Wynhoff, I., d'Ettorre, P., Leroy, C., Kajzer-Bonk, J., Maák, I. E., Barbero, F., Casacci, L. P., & Witek, M. (2025). Reintroduction provides insight into multimodal mimicry evolution in an ant social parasite. *XIX Congress of the Spanish Society of Ethology and Evolutionary Ecology*, *Seville*, Spain (Talk).

**Sánchez-García, D.**, Cerdá, X., & Angulo, E. (2024). Modificación del hábitat como herramienta de control de la invasión de la hormiga argentina. *XVIII International Congress of Myrmecology, Taxomara 2024*, *Málaga*, Spain (Talk).

**Sánchez-García, D.**, Cerdá, X., & Angulo, E. (2024). Recovering native ant communities by removing suitable conditions for the invasive Argentine ant. *IV International Young Researchers Conference on Invasive Species (IyrCIS)*, Online (Talk).

**Sánchez-García, D.**, Casacci, L. P., Wynhoff, I., Kajzer-Bonk, J., Sztencel-Jabłonka, A., Nowicki, P., & Witek, M. (2023). Changes in morphology and genetic structure in two populations of Phengaris (= Maculinea) teleius 30 years after separation. *Butterfly Conservation Symposium 2023*, *Wyboston*, England (Talk).

**Sánchez-García, D.**, Casacci, L. P., Wynhoff, I., Kajzer-Bonk, J., Nowicki, P., & Witek, M. (2022). Short-time evolution in the morphology of the myrmecophilous Maculinea teleius butterfly. *8th Polish Evolutionary Conference*, *Toruń*, Poland (Talk).

**Sánchez-García, D.** (2019). Fotografía de insectos: material, técnicas y aplicaciones. *XIV International Congress of Myrmecology, Taxomara 2019*, *Chefchaouen*, Morocco (Talk).

**Sánchez-García, D.**, Cerdá, X., & Angulo, E. (2018). ¿Temperatura o competencia, qué afecta más a las hormigas de Doñana? *XIII Iberian Congress of Myrmecology, Taxomara 2018*, *León*, Spain (Talk).

**Sánchez-García, D.**, Cuesta-Segura, A. D., Herraiz, J. A., Trigos-Peral, G., García García, F., Catarineu, C., Arcos González, J., & Fernández Martínez, J. A. (2017). Listado actualizado de las hormigas de la península ibérica e islas Baleares (Hymenoptera: Formicidae). *XII Iberian Congress of Myrmecology, Taxomara 2017*, *Madrid*, Spain (Poster).

**Sánchez-García, D.** (2016). Aspectos ecológicos sobre la mirmecocenosis de solana de la Sierra de Javalambre (Teruel). *XI Iberian Congress of Myrmecology, Taxomara 2016*, *Murcia*, Spain (Talk).

## Funding

| | | |
|---|---|---|
| 2022 - 2025 | PRELUDIUM grant 2021/41/N/NZ8/04360 (NCN, Poland). Bacteria – ant – plant interaction: the effect of ants as bacteria dispersal vectors and indirect drivers of plant health. | *202,001 PLN (≈ 46,000 €)* |

## Awards

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| Sep 2025 | Best oral presentation at the XIX International Congress of the Spanish Society of Ethology and Evolutionary Ecology (Seville, Spain) |

## Organization of Conferences and Scientific Events

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|---|---|---|
| 2021 | XVI International Conference of Myrmecology, Taxomara 2021 | *Organizing Committee and Scientific Committee* |
| 2020 | XV International Conference of Myrmecology, Taxomara 2020 | *Organizing Committee and Scientific Committee* |

## Reviews

Manuscripts reviewed for: Applied and Environmental Microbiology (1), Insect Conservation and Diversity (2), Arthropod-Plant Interactions (1), Polish Journal of Ecology (1)

## Workshops

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| 2025 | Invited participant in InvaPact III: analysis of the ecological effects of biological invasions |

## Courses

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| 2025 | Systematic Reviews and Meta-analysis in R |
| 2025 | Optimizing the use of R for ecological analyses |
| 2019 | Advanced University Course in Ecological Network Analysis |
| 2018 | Animal Tracking Course |

## Memberships

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|---|---|
| since 2025 | SORTEE - Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology |
| since 2012 | Iberian Association of Myrmecology (member of the Governing Board) |

# Skills

## Technical skills

| Coding languages | R packages and tools | Other |
|---|---|---|
| R | R Markdown / Quarto – ggplot2 – vegan – geomorph | GC-MS – Morphometrics – Network analysis – Population genetics – Ant taxonomy – Oxford Nanopore sequencing – Macro photography |

## Languages

| Skill | Spanish | English |
|---|---|---|
| Reading | Native | B2 |
| Writing | Native | B2 |
| Listening | Native | B2 |
| Speaking | Native | B2 |

Common European Framework of Reference for Languages: A1/A2: Basic User. B1/B2: Independent User. C1/C2: Proficient User

# References

- Postdoc: **Dra. Elena Angulo**, Doñana Biological Station - CSIC, Seville (Spain)
  angulo@ebd.csic.es
- PhD thesis: **Dra. Magdalena Witek**, Museum and Institute of Zoology, PAS, Warsaw (Poland)
  mwitek@miiz.waw.pl; mawitus@yahoo.co.uk
- Master's thesis: **Dr. Xim Cerdá**, Doñana Biological Station - CSIC, Seville (Spain)
  xim@ebd.csic.es
- Bachelor's thesis: **Prof. Dr. Ximo Baixeras**, University of Valencia, Valencia (Spain)
  joaquin.baixeras@uv.es